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***B.Tech. Degree I Semester Regular and Supplementary/
II Semester Supplementary Examination November 2018***

**CE/EE/ME/SE AS 15-1102 A & CS/EC/IT AS 15-1202 B ENGINEERING PHYSICS
(2015 Scheme)**

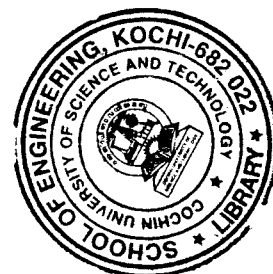
Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Differentiate between normal light and laser light.
- (b) How is the working principle of optical fibres based on total internal reflection?
- (c) Show that the atomic packing factor of SC crystal is 0.52.
- (d) What is meant by one-way shape memory alloy?
- (e) How are the nanoparticles prepared employing the top down methods?
- (f) Calculate the energy equivalent to 1 atomic mass unit in meV.
- (g) Why is uncertainty principle important for microscopic particles?
- (h) Write any four properties of ultrasonic waves.
- (i) Explain Einstein's photoelectric equation.
- (j) What are cooper pairs?



PART B

(4 × 10 = 40)

- II. (a) What you understand by holography? Explain its working and mention any four applications of holography. (6)
- (b) The coherence length of sodium light is 2.945×10^{-2} m and its wavelength is 5890 Å. Calculate the frequency and the coherence time. (4)

OR

- III. (a) What are optical fibres? Discuss the effect of dispersion on their performance. (6)
- (b) The NA of an optical fibre with cladding refractive index 1.59 is 0.2. Calculate the acceptance angle for the fibre in water. (4)
- IV. (a) What are Miller indices? How will you determine the Miller indices of a given plane? What are the distinct features of Miller indices? (6)
- (b) The density of a rock salt crystal is 2.16×10^3 Kg/m³. The interplanar spacing d_{100} is 2.82 Å. Determine the molecular weight of rock salt crystal. (4)

OR

- V. (a) Define metallic glasses. Explain how metallic glasses are prepared and how they are different from their crystalline counterparts. (6)
- (b) In a simple cubic, find the ratio of inter planar spacing of (1 1 0) and (1 1 1) planes. (4)

(P.T.O.)

- VI. (a) What are the different types of superconductors? Explain in detail. (6)
(b) The critical values of magnetic fields are 2×10^5 A/m and 10^5 A/m for niobium at 0 K and 8 K respectively. Determine its critical temperature. (4)

OR

- VII. (a) State and explain the fundamental postulates of special theory of relativity and derive Lorentz space time transformation equations on their basis. (6)
(b) In a laboratory, the life time of the particle moving with speed 2.8×10^8 m/sec is found to be 2×10^7 sec. Calculate the proper life time of the particle. (4)

- VIII. (a) Derive an expression for Schrödinger time dependent wave equation. (6)
(b) Calculate the smallest possible uncertainty in position of an electron moving with a velocity 3×10^7 m/sec. (4)

OR

- IX. (a) Describe the construction and working of piezoelectric generator to produce ultrasonic waves. (6)
(b) An ultrasound pulse sent by a source in sea is reflected by a submerged target at a distance 597.50 m and reaches the source after 0.83 seconds. Find the velocity of sound in sea water. (4)
